AI in Retail & E-commerce

Applying Artificial Intelligence to poor inventory management, low customer retention, and an inefficient supply chain.

Project Report

Submitted by

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Introduction

In retail and e-commerce, AI is used to monitor stock management, identify consumer demands, and deliver goods based on demands. AI can execute certain operations such as image and voice recognition, information perception, data learning, and decision-making.

The type of AI used by such companies is the example of Artificial Narrow Intelligence (ANI) since AI is developed for the execution of specific tasks. AI is neither AGI nor ASI. Such systems can apply reactive AI and limited memory AI whereas theory of mind and self-aware AI cannot be used.

The data becomes information once it is put in the right context. The data can be of different types depending on its format, that is, structured, semi-structured, or unstructured data. Structured data consists of sales figures in tabular form while semi-structured data includes JSON and event logging. Unstructured Data includes customer comments, pictures, and communications.

Poor Inventory Management

Problem

Oversupply is an ineffective management practice as it increases cost through locking in funds on the stock while shortage results into lost sales and dissatisfied customers. The demand is affected by seasonality, price, promotion, place, and life cycle of the product.

AI solution.

For predicting stockouts, supervised learning may be applied, and for estimating sales and predicting future demand, regression and time series forecast may be considered.

Feature engineering includes creation of useful features like holidays, promotion, price of the product, delivery time, and sales history. Unsupervised learning can also be used for the purpose of clustering of the products based on the demand behavior of each product.

Optimization takes into account predicted demand, inventory level at present, delivery time, budget constraints, and required service level for suggesting quantity of inventory to order and for transferring between stores. Reinforcement learning approach can also be taken into consideration for solving this problem.

Reasoning.

Bayesian inference updates risk assessment calculations with fresh data concerning sales or vendors. The process of deduction relies on certain rules concerning inventory, whereas the process of induction identifies any existing pattern within past sales. Abduction involves taking into account the most viable explanation regarding the unexpected shortage of inventory. Fuzzy logic can deal with such vague concepts as "high demand".

Expected Result.

Improving forecasting and inventory control may help to decrease both stockouts and overstocking of goods. In order to evaluate the efficiency of the system, its performance should be compared against the current system.

Low Customer Retention

Problem

Customers might not visit again if they get irrelevant offers, recommendations that do not suit them, and poor customer service. All customers have individual purchasing behavior, spending habits, and preferences as well as the likelihood of churning from the company.

AI Solution

Classification algorithm can be employed to predict the likelihood of churn customers. K-means clustering clusters customers according to their similarity of behavior without predefined categories.

Recommendation systems offer products to customers basing on their interests and preferences as well as those of customers who are similar to each other. NLP algorithms can be employed for topic modeling of customers' reviews and support emails. Bayesian reasoning enables the updating of the probability of churning given some new pieces of information.

Several useful parameters based on purchase frequency, total purchase, product preference, returns, and customer support.

Training and Evaluation

The dataset is divided into training, validation, and test datasets. If a fitted model has perfect accuracy on the training data, but poor accuracy on new data, then it faces overfitting, and underfitting occurs when the model is unable to identify useful features.

A confusion matrix, precision, recall, and F1 score will be used to evaluate the performance of the model. Accuracy alone may not give the correct picture when very few customers leave. Retention programs should also be tested to see if they are actually working.

Expected Result

The use of AI will allow businesses to identify those customers that need more focus and offer them better deals. In this way, the customers will be satisfied and will keep coming back to the business.

Inefficient Supply Chain

Problem.

Supplier delays, poor delivery routes, misuse of resources, and damage to the package are factors that can lead to high costs and unreliable deliveries.

State-Space Search

The problem may be described as follows: it has state space which includes the initial state, the goal state, the actions set, the transition function and path cost.

Breadth First Search is an algorithm that expands levels of nodes step by step. BFS algorithm uses a queue and finds the shortest paths in the unweighted graph.

Depth First Search expands one branch till its end before expanding others. The DFS algorithm uses a stack and typically requires less memory than BFS algorithm, but it does not find the shortest paths.

Best First Search algorithm uses heuristics for choosing the next node to expand. It is faster than the previous algorithms, but it does not provide the optimal solution.

A* Search uses the cost of the path and the estimated cost of reaching the goal as the components of its evaluation function.

$$f(n) = g(n) + h(n)$$

g(n) stands for the real cost from the start to the node n, whereas h(n) is the estimated cost from n to the goal node.

A* is optimal under certain conditions.

Prediction and perception.

Regression algorithms can be used to predict delivery time and cost, and predictive analytics is used to detect possible delays in deliveries. Optimization algorithms are used for vehicle assignment and routing.

Computer vision technology relies on CNNs (Convolutional Neural Networks) to analyse images of packages and detect damage that could be seen on them. Neural networks include artificial neurons organized in layers. The process of backpropagation changes the weights during the learning process. RNNs (Recurrent Neural Networks) may also be used if there is a need for processing sequential data.

Expected Result

The system may prove useful in minimizing delivery delays, optimal route planning, and quick package inspections. The following KPIs need to be taken into account: delivery time, delivery cost, delivery delay, and package damage rate.

Model Development and Evaluation

Identifying the business problem and baseline, gathering and preparing the data, generating features, dividing into training, validation, and testing, building a simple baseline model, training the simple model and other candidate models, tuning with the help of validation data, evaluating by using the held-out test data, conducting pilots, monitoring errors and drift, and updating if necessary.

For problems of regression and forecast – MSE, RMSE, absolute error measures, or relative error measures. The confusion matrix, precision, recall, F1 score, and accuracy apply to classification problems. Clustering is used for evaluation of stability and business relevance.

Business Key Performance Indicators (KPIs): Stock Out Ratio, Retention Ratio, Repeat Purchase Ratio, Punctuality Ratio, Time of Delivery, and Cost Per Order.

- · Transformers use self-attention algorithm and may be considered as members of encoder, decoder, and encoder-decoder architectural types.
- · Large language models use licensed data sources to summarize customer feedback or answer natural language queries. Large language models utilize authorized data sources for summarizing customer feedback or answering natural language queries. Diffusion models produce data, such as images. Example of multimodal AI is one where text, image, and sound are combined together.
- There are actions such as task planning and using tools which require permissions, logging, action approvals, and backup plans. It is just the extra features which are not necessary in every case.
- Ethics comprises of principles of fairness, transparency, accountability, privacy, and safety.
- Bias by data, algorithm, and labelling can affect the result; obscure algorithms can create the black box issue.
- Customer data has to be kept safe; unnecessary data should not be gathered, there should be limitations, human review should be done in case of important decisions.
- Considering the EU AI Act and the Indian AI Safety Recommendations, wherever applicable.
- The use cases with common foundations include health prediction and imaging, quality control in manufacturing, predictive maintenance, financial fraud prevention, and educational help.
- The India AI mission of India and Safe & Trusted AI area are important in relation to our use case.

Conclusion

Inventory Management is associated with Supervised Learning, Forecasting, Reasoning and Optimization; Retention is associated with Classification, Clustering, Recommendations, Natural Language Processing, and Evaluation; Efficient Supply Chain Management is associated with Search, Prediction, Optimization, and Computer Vision.

Generators and agents based on AI might be helpful in some cases, but for their effective use proper data and metrics must be available.

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Appendices

Appendix A. Inventory Management

Table A1. Techniques and applications

Techniques/Concepts	Applications
Supervised Learning/Regression	Prediction of stock-outs and sales quantities.
Time-Series Forecasting	Future demand forecasts using trend and seasonality whenever appropriate.
Feature Engineering	Past variables for sales, promotions, holidays, prices, and lead time.
Optimization	Propose quantities to be reordered or transferred.
Reinforcement Learning	Adaptive replenishment evaluation through simulation before real implementation.
Bayesian Reasoning and Logic	Update estimations and re-evaluation through changing information.

Figure A1. Illustrative demand vs forecast

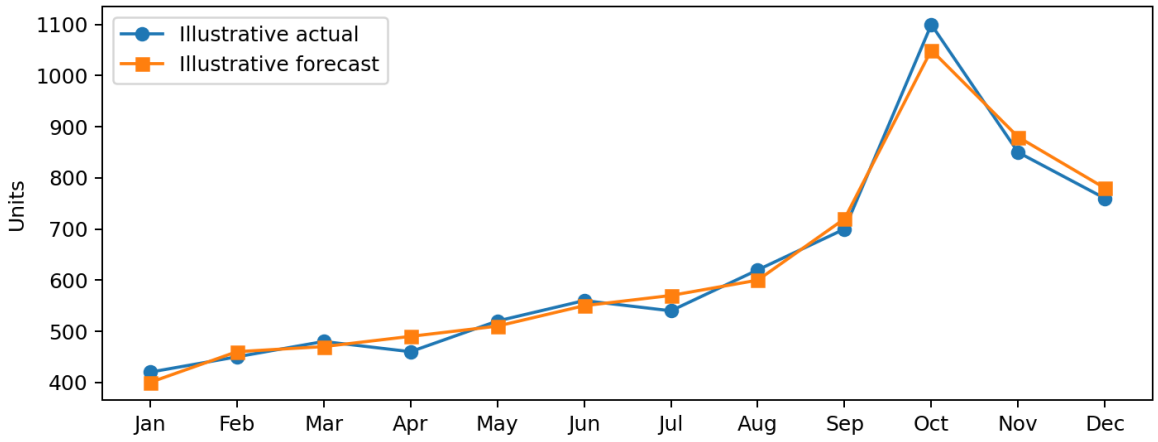


Figure A1. Synthetic illustrative values, not retailer data.

Figure A2. Inventory workflow

Stage	Description
Data	Inventory, sales, promotion, pricing, vendor lead times; structured and event data.
Prepare/Predict	Record preparation, feature engineering, forecasting demand and out-of-stock probabilities.
Optimize/Reason	Re-estimate and re-optimize replenishments or stock transfer.
Monitor	Monitor forecasting accuracy and KPIs.

Appendix B. Customer Retention

Table B1. Techniques and applications

Technique / concept	Application
Classification	Classify for churn and evaluate using confusion matrix, precision, recall, F1.
K-Means	Classification to form customer clusters without labels.
Recommendation systems	Collaborative, content-based, or hybrid.
NLP/generative AI	Classify and summarize customer reviews.
Bayes Theorem	Update probability of churn as evidence is added.
Feature engineering and data splitting	Recency/frequency/value features; train/validation/test.

Figure B1. Synthetic customer segments

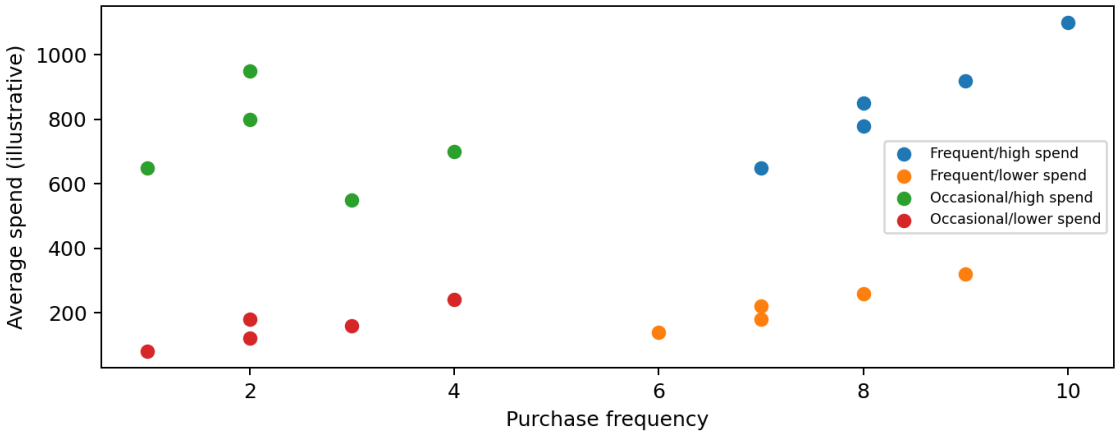


Figure B1. Synthetic segments only; not real customer data or trained K-Means output.

Figure B2. Retention workflow

Stage	Description
Data	Purchase history, interactions, service contacts, feedback with appropriate permissions.
Model	Estimate churn, segment shoppers, rank products, summarize feedback.
Action	Test relevant service interventions or offers with oversight.
Evaluate	Measure retention, repeat purchase, complaints, and campaign impact.

Appendix C. Supply-Chain Efficiency

Table C1. Techniques and applications

Technique / concept	Application
Predictive modelling / regression	Forecasting of delivery time and mark any delays.
State space representation	States, initial state, goal state, actions, transition function, and path cost.
Breadth-first search / depth-first search	Breadth-first search is an optimal shortest path for unweighted graph problems; depth-first search uses less memory but is not optimal.
Greedy Best-First search	greedy algorithm based on heuristic and is fast but not necessarily optimal.
A* Algorithm	f(n) = g(n) + h(n).
Optimization / Reinforcement learning	Resource plan and policy testing with protection.
Computer vision / CNN	Analyze packages photos for defects.

Figure C1. Supply-chain workflow

Stage	Description
Inputs	Orders, supplier notifications, warehousing activities, logistics information, package images.
Predict	Predict delivery time and possibility of delay.
Search/optimization	Route comparisons and resource allocation within restrictions.
Inspect/monitor	Locate packages for inspection and evaluate delivery performance metrics.

Figure C2. Illustrative supply-chain KPIs

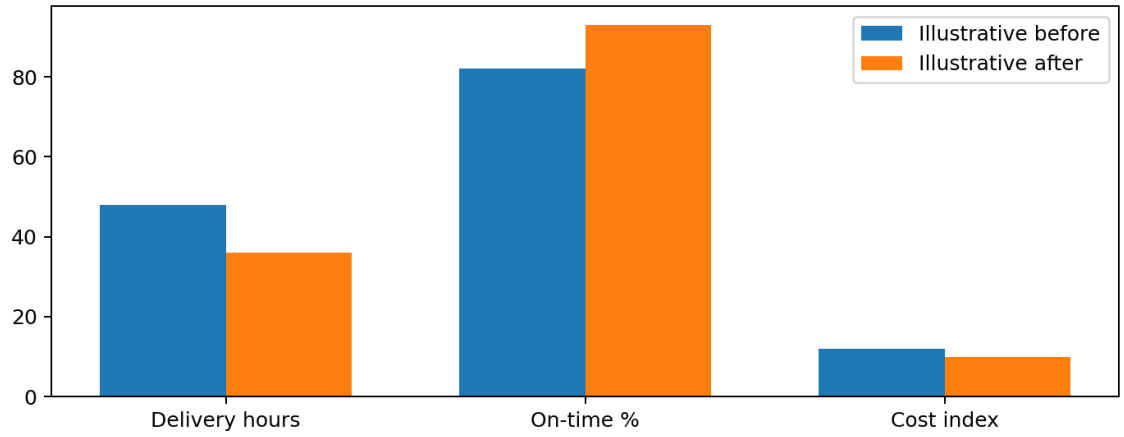


Figure C2. Illustrative values only, not measured results or guarantees.

Appendix D. KPI and Model Evaluation Table

Area	Metric	Meaning	Direction
Inventory	Stockout Probability	Percentage of intervals during which there is no stock	Lower
Inventory	Forecast Error (MAE, RMSE, WAPE)	Difference between forecasted and actual demand	Lower
Inventory	Inventory Holding Cost	Additional costs from holding excess stock	Lower
Retention	Churn Probability	Probability that a customer meets the criteria for inactivity or churn	Lower
Retention	Purchase Frequency or Retention Rate	Rate of customer purchases or retention within a group	Higher
Classification	Confusion Matrix	Shows true positives, false positives, true negatives, and false negatives	See all
Classification	Precision, Recall, F1 Score	Measures classification performance and the balance between precision and recall	Optimal balance
Regression	MSE or RMSE	Measures numerical forecasting error	Lower
SCM	Delivery Delay or On-Time Delivery	Delivery time and percentage of orders delivered on time	Lower delivery time and higher on-time delivery
SCM	Cost per Order	Average cost of fulfilling and transporting an order	Lower without reducing reliability
Inspection	Defect Classification	Precision and recall for validated package images	High defect detection and low false positives

Evaluation checklist

- Establish a baseline and define KPIs before deployment.
- Avoid data leakage; use suitable train/validation/test splits and check overfitting/underfitting.
- Report sample size, measurement window, false positives and false negatives where relevant.
- Treat all charts and example values as illustrative, not proof of impact.

Appendix E. Responsible AI checklist

- Confirm purpose, permissions, access controls, and data retention.
- Check data, algorithmic, and label bias; consider fairness and explainability.
- Define human review, escalation, fallback, and accountability.
- Monitor safety, privacy, drift, and real business outcomes.
- Review relevant legal/policy guidance, including Indian guidance and the EU AI Act where applicable.
- Scale only when evidence supports the cost, risk, and readiness.